

UNIVERSITY OF EDINBURGH
COLLEGE OF SCIENCE AND ENGINEERING
SCHOOL OF INFORMATICS

INFR09027 COMPUTER COMMUNICATIONS AND NETWORKS

Friday 23rd August 2013

09:30 to 11:30

INSTRUCTIONS TO CANDIDATES

Answer any TWO questions.

All questions carry equal weight.

CALCULATORS MAY NOT BE USED IN THIS EXAMINATION

Year 3 Courses

Convener: K. Kalorkoti
External Examiners: K. Eder, T. Field

THIS EXAMINATION WILL BE MARKED ANONYMOUSLY

1. (a) List two advantages and two disadvantages of having international standards for network protocols. Name two prominent sets of standards relevant to computer networking. [6 marks]
- (b) Given that each network interface at a router or host is assigned a unique IP address, why is a separate MAC address (also called physical address) needed? [2 marks]
- (c) Datagram networks route each packet as a separate unit, independent of all others. This is not the case with virtual-circuit networks, where each data packet follows a predetermined route. Does this observation imply that virtual-circuit networks cannot route isolated packets from an arbitrary source to an arbitrary destination? Explain. [2 marks]
- (d) Consider a packet-switched network in which data of size z bits is transmitted as a series of packets over a k -hop path. Each packet contains d data bits and h header bits, where the combined size of the header and data bits, $(d + h)$, in each packet is much smaller than the overall size of the transmitted data (z). What is the value of d which minimises the total delay, assuming that the bit rates of the lines are uniformly b bits/sec (you may assume that the propagation delay is negligible)? [7 marks]
- (e) The high-quality version of an hour-long TV programme on the BBC iPlayer has a size of 350 Mbytes to download onto a home computer. A user has a broadband service with a download speed of 2Mbps. The round-trip time to the BBC iPlayer website is determined to be 100ms by using a 'ping' test with a default packet size of 56bytes. (You should assume that the access link is the bottleneck and delays other than those due to propagation and transmission are negligible.) Calculate the time taken for the user to download the entire programme and the throughput of the transfer. How much does the propagation delay matter for this transfer? [8 marks]

2. (a) List the key differences between IPv4 and IPv6. [5 marks]
- (b) Describe two mechanisms for transition from IPv4 to IPv6. [4 marks]
- (c) Give four key differences between circuit and packet switching. [4 marks]
- (d) A system has an n -layer protocol hierarchy. Applications generate messages of length M bytes. At each of the layers, an h -byte header is added. What fraction of the network bandwidth is filled with headers? [3 marks]
- (e) Assume that TCP implements an extension that allows window sizes much larger than 64 KB. Suppose that you are using this extended TCP over a 1-Gbps link with a round-trip propagation delay of 100 ms to transfer a 10-MB file, and the TCP receive window is 1 MB. If TCP sends 1-KB packets (assuming no congestion and no lost packets):
- i. How many round-trip times (RTTs) does it take until slow start opens the send window to 1 MB? [2 marks]
 - ii. How many RTTs does it take to send the file? [3 marks]
 - iii. If the time to send the file is given by the number of required RTTs multiplied by the RTT value, what is the effective throughput for the transfer? What percentage of the link bandwidth is utilized? [4 marks]

3. (a) What is CIDR (Classless InterDomain Routing)? What is the impact of using CIDR on forwarding? [4 marks]
- (b) What does pipelining mean in the context of reliable data transfer protocols? Describe two well-known pipelined reliable data transfer protocols. Give an example of an application layer protocol in which a pipelining-like approach is used. [4 marks]
- (c) BGP and RIP are Internet routing protocols that rely on TCP or UDP for their message exchanges. Since they are conceptually part of the network layer, why do they not just use raw IP packets? [3 marks]
- (d) An object of size S needs to be transferred over a 2-Mbps long-distance link with a round-trip propagation delay of 500 ms. At what value of the transfer size S will the transmission delay start to dominate in the overall transfer delay. Assume delays other than due to propagation and transmission to be negligible. [6 marks]
- (e) Two CSMA/CD stations are each trying to transmit long (multiframe) files. After each frame is sent, they contend for the channel, using the binary exponential backoff algorithm. What is the probability that the contention ends on round k , and what is the mean number of rounds per contention period? [8 marks]